Midterm Laboratory Practical Exam

Problems:

1. **Is 3-digit grouping prime?**

   The input is any positive integer greater than 100. The output is the string "prime" or "composite" for every 3-digit grouping of the input integer starting from the ones digit. Example:

   ```
   12997
   prime
   composite
   ```

   Explanation:

   12-composite 997-prime

2. **Self-powers to literal text**

   The series:

   $$1^2 + 2^2 + 3^2 = 14$$

   Is a self-power. Ask for an input within 2<n≤9, then display the self-powers result in integer format and in digit character format (i.e. each digit corresponds to a letter in the alphabet e.g. 0-A, 1-B, 2-C)

   Example:

   ```
   4
   30
   DA

   //Explanation:
   //4: 1^2 + 2 ^ + 3^2 + 4^2 = 30, 3-D 0-A
   ```

   Hint: character value for 'A' is 65, 'B' is 66, 'C' is 67…

3. **Combined to create palindromes**

   A palindrome is a sequence of characters that would read the same when read from left to right or right to left for example:

   ```
   1221 is a palindrome
   ```

   Ask 2 integer input, that when concatenated, it would not exceed 8 digits. Example:

   ```
   A: 133
   B: 112
   Concatenated: 133112
   ```

   Hint: 133 * 10 ^ digit Count of B (i.e. 3) =133000 + B = 133112

   Check if the concatenated digits A and B is palindrome or not. Example:

   ```
   A: 1334
   B: 331
   Concatenated: 1334331
   Palindrome
   ```

4. Composite Sum of Rotated Digits

Ask an integer input with digit *d* within 2≤*d*≤7. Rotate the digits then for each new value created from the rotation find the sum if it's a composite number.

Example:

```
Input: 937
Composite Sum: 0
Rotation: 793
Composite Sum: 793
Rotation: 397
Composite Sum: 793
```

5. Largest Product of Truncated Digits

The number 123503 would produce different digit(s) products when continuously removing digits from left to right,
123503: 1 x 2 x 3 x 5 x 0 x 3 = 0,
23503: 2 x 3 x 5 x 0 x 3 = 0,
3503: 3 x 5 x 0 x 3 = 0,
503: 5 x 0 x 3 = 0,
03: 0x3=0,
3: 3=3

and from right to left,
123503: 1x2x3x5x0x3 = 0,
12350: 1x2x3x5x0 = 0,
1235: 1x2x3x5 = 30,
123: 1x2x3 = 6,
12: 1x2 = 2,
1: 1 = 1

Your input is a positive integer with digits *d* within 2≤*d*≤8, the output is the largest products among the truncated digits. Example:

```
142012
8
```